

GATE 2026 15TH FEB 2026 S2

Candidate ID	DA26S86210680
Candidate Name	ADITYA KUMAR JHA
Test Center Name	ION Digital Zone iDZ SARONA
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Subject	DA Data Science and Artificial Intelligence

Section : **General Aptitude****Q.1**

'If his latest movie had been a commercial success, the actor would have made enough money to sponsor his next movie.'

Based only on the above sentence, which one of the following statements is true?

Options A.

The actor will certainly sponsor his next movie.

B.

His latest movie was not commercially successful.

C.

The actor made enough money from his latest movie.

D.

His latest movie was a commercial success.

Question Type : **MCQ**Question ID : **22848210015**Status : **Answered**Chosen Option : **B**

Q.2

The four pieces of a puzzle are shown in the figure below.



Which one of the figures labelled as P, Q, R, and S can be constructed by using each of the four pieces only once without overlaps?



P



Q



R



S

Options A.

Q

B.

R

C.

P

D.

S

Question Type : MCQ
Question ID : 22848210013
Status : Answered
Chosen Option : C

Q.3

Verbosity : Brevity :: Insolence : _____

Choose the word that best fills the blank.

Options A.

Innocence

B.

Wealth

C.

Respect

D.

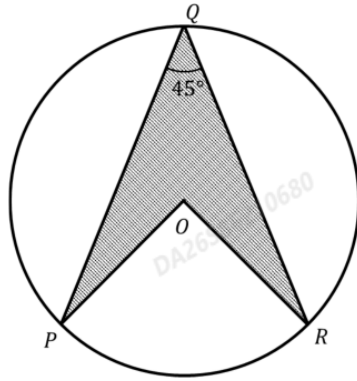
Solace

Question Type : **MCQ**Question ID : **22848210011**Status : **Not Answered**

Chosen Option : --

Q.4

In the given figure, P , Q , and R are three points on a circle of radius 10 cm with O as its center, $\overline{PQ} = \overline{RQ}$, and $\angle PQR = 45^\circ$. The figure is representative. The area of the shaded region $PQRO$ is _____ cm^2 .



Options A.

50 $\sqrt{2}$

B.

50

C.

100

D.

25 $\sqrt{2}$

Question Type : MCQ

Question ID : 22848210020

Status : Not Attempted and
Marked For Review

Chosen Option : --

Q.5

The product of the digits of a three-digit number is 70. The sum of the digits of this three-digit number is _____

Options A.

14

B.

16

C.

18

D.

12

Question Type : MCQ

Question ID : 22848210012

Status : Answered

Chosen Option : A

Q.6

Five integers are picked from 0 to 20, with possible repetitions, such that their mean is 12, median is 18, and they have a single mode of 20.

Ignoring permutations, the number of ways to pick these five integers is _____

Options A.

2

B.

3

C.

0

D.

1

Question Type : MCQ

Question ID : 22848210017

Status : Answered

Chosen Option : C

Q.7

Consider two distinct positive real numbers m, n , with $m > n$.

Let $x = n^{\log_{10}(m)}$ and $y = m^{\log_{10}(n)}$. The relation between x and y is _____.

Options A.

A.
 $x = \log_{10}(y)$

B.

B.
 $x > y$

C.

C.
 $x = y$

D.

D.
 $x < y$

Question Type : MCQ

Question ID : 22848210014

Status : Answered

Chosen Option : B

Q.8

P , Q , R , S , X , and Y are distinct single-digit whole numbers taking values from 0 to 9.

PQ is a two-digit number with Q being in the units place and P in the tens place. Similarly, RS is a two-digit number.

It is known that PQ and RS are consecutive numbers and $(PQ)^2 + (RS)^2 = XYP$, with XYP being a three-digit number.

The value of Y is _____

Options A.

4

B.

6

C.

7

D.

5

Question Type : MCQ

Question ID : 22848210019

Status : Answered

Chosen Option : B

Q.9

'My friend and I parted ___ the door ___ the cabin that I had rented ___ the night.'

Choose the option with the correct sequence of words to fill the blanks.

Options A.

of; for; in

B.

in; of; for

C.

at; of; for

D.

for; at; of

Question Type : **MCQ**Question ID : **22848210016**Status : **Answered**Chosen Option : **B**

Q.10

Rishi and Swathi are students of Class 5. Pavan and Tanvi are students of Class 4. Rishi and Pavan are boys. Swathi and Tanvi are girls. The four students played a total of three games of chess. The games were played one after another. A player who lost a game did not participate in any more games. It was observed that:

- (i) the first game was the only game where two students of the same class played against each other,
- (ii) the students of Class 5 won more games than the students of Class 4, and
- (iii) the boys won two games and the girls won one game.

The student who did not lose any game is _____.

Options A.

Tanvi

B.

Pavan

C.

Rishi

D.

Swathi

Question Type : MCQ

Question ID : 22848210018

Status : Answered

Chosen Option : A

Section : DA Data Science and Artificial Intelligence

Q.1

Suppose that a computer program provides a non-negative and integer-valued random solution to the equation $n_1 + n_2 + n_3 + n_4 = 20$.

Which of the following is the probability that all of n_1, n_2, n_3, n_4 in the provided solution are positive?

Options

A. $\binom{19}{3} / \binom{23}{3}$

B. $\binom{19}{4} / \binom{24}{4}$

C. $\binom{20}{4} / \binom{24}{4}$

D. $\binom{20}{3} / \binom{23}{3}$

Question Type : MCQ

Question ID : 22848210030

Status : Not Answered

Chosen Option : --

Q.2

Let M be a randomly chosen non-empty subset of $S = \{1, 2, 3, \dots, 2026\}$.

Which of the following is the probability that the product of all the elements of M is even?

Options A.

$2^{1013}(2^{1013} - 1)/2^{2026}$

B.

$2^{1013}/2^{2026}$

C.

$1/(2^{2026} - 1)$

D.

$2^{1013}(2^{1013} - 1)/(2^{2026} - 1)$

Question Type : MCQ

Question ID : 22848210029

Status : Not Answered

Chosen Option : --

Q.3

Consider that for a supervised learning task, the objective function being minimized is $f_w(x) = wx$, where $x \in \mathbb{R}$ is the input and $w \in \mathbb{R}$ is the parameter. Stochastic Gradient Descent with learning rate of 0.10 is used for parameter updates.

Suppose that at the end of iteration i , the value of w becomes 10.00.

Let $x = 10.00$ be the input for iteration $(i + 1)$.

The value of w at the end of iteration $(i + 1)$ is _____. (Rounded off to two decimal places)

Given **20.00**

Answer :

Question Type : **NAT**

Question ID : **22848210039**

Status : **Marked For Review**

Q.4

Sentence X is said to entail Sentence Y if whenever X is *TRUE*, Y also must hold *TRUE*.

Which of the following statements is/are correct if X entails Y ?

Options A.

$X \Rightarrow Y$

B.

if Y then X

C.

$X \wedge \neg Y$ is *FALSE*

D.

if X then Y

Question Type : **MSQ**

Question ID : **22848210034**

Status : **Marked For Review**

Chosen Option : **A,D**

Q.5

Let $M = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$ be a 2×2 matrix, where $\theta = \frac{2\pi}{5}$, and

$$I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Which of the following options is equal to M^{2026} ?

Options A.

$$I_2$$

B.

$$M^{-1}$$

C.

$$M^2$$

D.

$$M$$

Question Type : MCQ

Question ID : 22848210031

Status : Not Attempted and Marked For Review

Chosen Option : --

Q.6

Consider that you are training a classifier for a 10-class classification problem. Each input is represented as a 512-dimensional vector. There are 1000 samples, out of which first 100 will be used for testing. Let *Leave-One-Out-Cross-Validation* (LOOCV) be used for selection of the classifier model before testing.

Which of the following options is the correct number of validation splits that will be generated?

Options

A. 1000

B. 512

C. 900

D. 10

Question Type : MCQ

Question ID : 22848210022

Status : Not Answered

Chosen Option : --

Q.7

Consider two relations r and s defined on the relational schemas $R(A, B)$ and $S(E, C)$, respectively. A is the primary key of R and E is a foreign key of S referencing A in R .

Which of the following operations will **NEVER** violate the foreign key constraint?

Options A.

Deleting records from relation r

B.

Inserting records into relation s

C.

Deleting records from relation s

D.

Inserting records into relation r Question Type : **MSQ**Question ID : **22848210036**Status : **Answered**Chosen Option : **A,B,C**

Q.8

The value of $\sum_{i=0}^{\infty} \sum_{j=1}^{\infty} 2^{-i} 3^{-j}$ is _____. (Answer in **integer**)

Given 1
Answer :

Question Type : **NAT**Question ID : **22848210045**Status : **Answered**

Q.9

Consider that the quick sort algorithm is used to sort an array of n distinct randomly ordered elements. In every call, the pivot is chosen as the first element of the current subarray.

Let $T(n)$ denote the expected time to sort the array. Assume that the time to partition is linear in the size of the current subarray.

Which of the following recurrence relations correctly represents $T(n)$ in this scenario?

Options

A. $T(n) = T\left(\frac{n}{4}\right) + T\left(\frac{3n}{4}\right) + O(n)$

B. $T(n) = 2T\left(\frac{n}{2}\right) + O(n)$

C. $T(n) = T(1) + T(n-1) + O(n)$

D. $T(n) = \frac{1}{n} \sum_{k=0}^{n-1} [T(k) + T(n-k-1)] + O(n)$

Question Type : MCQ

Question ID : 22848210025

Status : Not Answered

Chosen Option : --

Q.10

Which of the following algorithms is **NOT** an example of uninformed search?

Options A.

A* Search

B.

Depth-limited Search

C.

Breadth First Search

D.

Depth First Search

Question Type : MCQ

Question ID : 22848210023

Status : Answered

Chosen Option : A

Q.11

Suppose a random variable Z follows $Normal(\mu = 0, \sigma^2 = 1)$ distribution with probability density function $g(z)$ and cumulative distribution function $G(z)$. Another random variable Y follows t_1 distribution with probability density function $h(y)$ and cumulative distribution function $H(y)$. Let c be the positive real number for which $g(c) = h(c)$.

Which of the following statements is/are correct?

Options A.

A. $g(0) = h(0)$

B.

$G(0) = H(0)$

C.

$G(-c) < H(-c)$

D.

$G(c) < H(c)$

Question Type : **MSQ**

Question ID : **22848210038**

Status : **Answered**

Chosen Option : **A,D**

Q.12

In the following table, the **Task** column lists a few tasks related to machine learning. The **Algorithm** column lists a few algorithms.

Each entry “t” from the **Task** column is to be matched with an appropriate entry “a” from the **Algorithm** column such that the task “t” can be solved using the algorithm “a”. Denote such a match as t:a

Task	Algorithm
T1 – Clustering	A1 – Markov Chain Monte Carlo
T2 – Classification	A2 – K-Medoid
T3 – Sampling	A3 – Linear Discriminant Analysis
T4 – Feature Extraction	A4 – Naive Bayes

Which of the following options is/are the correct matching(s)?

Options A.

T1:A2, T2:A4, T3:A1, T4:A3

B.

T1:A4, T2:A2, T3:A1, T4:A3

C.

T1:A4, T2:A3, T3:A1, T4:A2

D.

T1:A3, T2:A4, T3:A1, T4:A2

Question Type : **MSQ**

Question ID : **22848210033**

Status : **Answered**

Chosen Option : **A**

Q.13

Consider that the visualization of a 3-dimensional data cube is showing *Sales Quantity* for each combination of the attributes *Product Type*, *Month* and *Country*.

From this, if we want to further visualize the *Sales Quantity* for each combination of *Product Type*, *Month* and *State*, which of the following OLAP operations should be performed?

Options A.

Slicing

B.

Drill-down

C.

Roll-up

D.

Dicing

Question Type : MCQ

Question ID : 22848210028

Status : Marked For Review

Chosen Option : B

Q.14

You are given the following Pre-order and In-order traversals of a Binary Tree T with nodes E, F, G, P, Q, R, S.

Pre-order: P Q S E R F G

In-order: S Q E P F R G

Which of the following statements is/are true about the Binary Tree T?

Options A.

Node Q has only one child

B.

The left subtree of node R contains the node G

C.

Node P is the root of T

D.

The Post-order traversal of T is:

S E Q F G R P

Question Type : **MSQ**

Question ID : **22848210035**

Status : **Marked For Review**

Chosen Option : **A,C**

Q.15

Which of the following statements is **NOT** true?

(The names of the predicates are intuitive.)

Options A.

"Each king is a person" is equivalent to $\forall x \text{ IsKing}(x) \wedge \text{IsPerson}(x)$

B.

$\forall x \forall y \text{ Classmate}(x, y) \Rightarrow \text{Classmate}(y, x)$

C.

$\forall x \text{ Likes}(x, \text{Icecream}) \Rightarrow \neg \exists x \neg \text{Likes}(x, \text{Icecream})$

D.

"All humans are mortal" is equivalent to $\forall x \text{ IsHuman}(x) \Rightarrow \text{IsMortal}(x)$

Question Type : MCQ

Question ID : 22848210024

Status : Marked For Review

Chosen Option : B

Q.16

Let A be a sorted array containing 1000 distinct integers. You perform a recursive binary search on A to find an element y. Suppose each comparison checks whether the middle element computed during the current recursive step is equal to, less than, or greater than y.

The maximum number of comparisons that may have to be performed if y is not an element of A is _____. (Answer in **integer**)

Given 10
Answer :

Question Type : NAT

Question ID : 22848210041

Status : Answered

Q.17

Consider a set $S_1 = \{x = (x_1, x_2, x_3)^T \in \mathbb{R}^3 \mid x^T x \leq 16\}$. Let S_2 be another set which is a subspace of $\mathbb{R}^3$ with dimension two.

Which of the following gives the area of $S_1 \cap S_2$?

Options A.

A. 4π

B.

16π

C.

$16\pi^2$

D.

$4\pi^2$

Question Type : MCQ

Question ID : 22848210032

Status : Not Answered

Chosen Option : --

Q.18

For a classification problem, Principal Component Analysis (PCA) has been used to reduce the dimensionality of a feature space from 100 to 10.

Which of the following options is true about the angle θ between the first and the tenth principal components?

Options A.

$\theta = 0^\circ$

B.

$\theta = 90^\circ$

C.

$0 < \theta < 90^\circ$

D.

$90^\circ < \theta \leq 180^\circ$

Question Type : MCQ

Question ID : 22848210021

Status : Not Attempted and
Marked For Review

Chosen Option : --

Q.19

Consider the given Python program.

```
def append_to_lst(val, lst=[]):  
    lst.append(val)  
    return lst  
print(append_to_lst(1))  
print(append_to_lst(2))  
print(append_to_lst(3, []))
```

Which of the following is the correct output of this program?

Options A.

[1]
[1, 2]
[3]

B.

[1]
[2]
[1, 2, 3]

C.

[1]
[2]
[3]

D.

[1]
[1, 2]
[1, 3]

Question Type : **MCQ**

Question ID : **22848210026**

Status : **Not Attempted and
Marked For Review**

Chosen Option : --

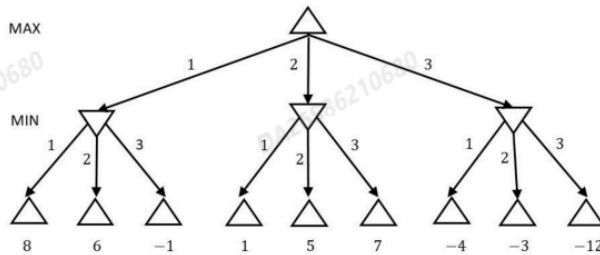
Q.20

Consider the game tree for a two-player turn-taking minimax game as shown in the figure. The value of a terminal node represents the utility of the game state if the game ends there. The numbers written next to the edges denote the strategies.

There are two players MAX and MIN. At any particular state of the game, MAX prefers to move to a state of maximum value. On the other hand, MIN prefers to move to a state of minimum value.

Suppose MAX starts the game at the root and has three strategies: 1, 2 and 3. Next, MIN plays and also has three strategies: 1, 2 and 3. The game ends there. Both players always take optimal strategies throughout the game.

At the root, the best strategy for MAX is _____. (Answer in **integer**)



Given 2

Answer :

Question Type : NAT

Question ID : 22848210040

Status : Answered

Q.21

Let X be an exponentially distributed random variable with mean $\lambda (> 0)$. If $P(X > 5) = 0.35$, then the conditional probability $P(X > 10 | X > 5)$ is _____. (Rounded off to **two** decimal places)

Given 0.00

Answer :

Question Type : NAT

Question ID : 22848210044

Status : Answered

Q.22

The number of bijections $f(\cdot)$ from the set $S = \{1, 2, 3, 4\}$ to itself such that $f(f(n)) = n$, for all $n \in S$, is _____. (Answer in **integer**)

Given 4

Answer :

Question Type : NAT

Question ID : 22848210043

Status : Answered

Q.23

Let $R(A, B, C, D, E)$ be a relational schema with functional dependency set $F = \{A \rightarrow BC, CD \rightarrow E, E \rightarrow A\}$.

Which of the following statements is correct?

Options

- A. **A and CD** are the only candidate keys of R.
- B. **AD, ED and CD** are the only candidate keys of R.
- C. **A, E and CD** are the only candidate keys of R.
- D. **AD and ED** are the only candidate keys of R.

Question Type : MCQ

Question ID : 22848210027

Status : Not Attempted and
Marked For Review

Chosen Option : --

Q.24

In a relational database, a B+ Tree Index is to be constructed for a relation on a key field. In a B+ Tree, a Node Pointer points to a sub-tree and a Data Record Pointer points to a block of database records.

Let, Node size = 4096 bytes, Node Pointer size = 10 bytes, Search Key Field size = 11 bytes and Data Record Pointer size = 12 bytes.

The maximum number of Node Pointers that can be present in a non-leaf node of the B+ Tree is _____. (Answer in integer)

Given 409

Answer :

Question Type : NAT

Question ID : 22848210042

Status : Answered

Q.25

Let $f(x) = x^3 - 3x^2 + 2$ be a function defined on $(-1, 3]$.

Which of the following statements is/are correct?

Options A.

A. $f(x)$ has a root at 1.

B.

B. $f(x)$ has exactly two roots in $[-0.9, 0]$.

C.

C. $f(x)$ has maximum at 0 only.

D.

D. $f(x)$ has a minimum at 2 only.

Question Type : MSQ

Question ID : 22848210037

Status : Marked For Review

Chosen Option : A

Q.26

Let X be a discrete valued random variable with cumulative distribution function $F(x)$.

Which of the following statements is/are correct?

Options

A. $F(x)$ is always a positive function.

B. $F(x)$ is a left continuous function.

C. $F(x)$ has jump discontinuity.

D. $F(x)$ is a non-decreasing function.

Question Type : MSQ

Question ID : 22848210064

Status : Marked For Review

Chosen Option : A,D

Q.27

Let $M = (I_n - \frac{1}{n} \mathbf{1}\mathbf{1}^T)$ be a matrix, where $\mathbf{1} = (1, 1, 1, \dots, 1)^T \in \mathbb{R}^n$ and I_n is the identity matrix of order n .

Which of the following options is/are correct?

Options A.

$\text{Trace}(M) = n$

B.

M is a projection matrix

C.

$M^T = M$

D.

$M^2 = I_n$

Question Type : **MSQ**

Question ID : **22848210062**

Status : **Answered**

Chosen Option : **A,D**

Q.28

Let four points in three-dimensional space be:

P1: [2, 3, -1], P2: [3, 1, 1], P3: [5, -2, 3] and P4: [3, 3, 3].

Hierarchical Agglomerative Clustering is used to cluster the above points. If Manhattan Distance is used as the distance metric during clustering, which of the following options indicates the two points that will be merged first?

Options A.

P2, P3

B.

P2, P4

C.

P1, P2

D.

P3, P4

Question Type : MCQ

Question ID : 22848210046

Status : Answered

Chosen Option : C

Q.29

Consider that Linear Ridge Regression is being used to learn a prediction function $y_{pred} = w^T x$, where $w, x \in \mathbb{R}^2$ and Mean Absolute Error (MAE) is used to measure the prediction error. A weight of 0.20 is associated with the regularizer.

At an intermediate step of the training process, assume that the parameter $w = [-3.00, 4.00]^T$. In the next step, for the input $x = [1.00, 2.00]^T$, the predicted value of y is noted. Let the relation between $x = [x_1, x_2]^T$ and the true value of y be $y_{true} = x_1 + x_2$.

The value of the overall regularized loss function for this instance is _____.
(Rounded off to two decimal places)

Given --
Answer :

Question Type : NAT

Question ID : 22848210065

Status : Not Attempted and
Marked For Review

Q.30

Let $\gamma_1, \gamma_2, \gamma_3$ be the eigenvalues of the matrix $\begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos t & \sin t \\ 0 & -\sin t & \cos t \end{bmatrix}$,

where $t \in [-\pi, \pi]$ is in radians.

Which one of the following options lists all the possible values of t satisfying $\gamma_1 + \gamma_2 + \gamma_3 = 1 + \sqrt{2}$?

Options A.

$\left\{ \frac{\pi}{4}, -\frac{\pi}{4} \right\}$

B.

$\left\{ \frac{\pi}{3}, -\frac{\pi}{4} \right\}$

C.

$\left\{ \frac{\pi}{3}, -\frac{\pi}{3} \right\}$

D.

$\left\{ \frac{\pi}{4}, -\frac{\pi}{3} \right\}$

Question Type : MCQ

Question ID : 22848210056

Status : Marked For Review

Chosen Option : A

Q.31

Consider the problem of sorting the given array in ascending order:

$P = [1, 2, 3, 5, 4]$

Consider two sorting algorithms Bubble Sort (BS) and Insertion Sort (IS).

Let $N1$ be the total number of comparisons done by BS on the elements of P and $N2$ be the total number of comparisons done by IS on the elements of P .

Which of the following options is/are correct?

Options

A. $N1 = 10, N2 = 4$

B.

Both BS and IS on P will make at least one unnecessary comparison (i.e., comparing elements that are already in correct order)

C. IS on P will perform only one swapD. $N1 > N2$

Question Type : MSQ

Question ID : 22848210059

Status : Answered

Chosen Option : C,D

Q.32

Consider the given relations X , Y and Z . The relation X has three columns P , Q and R . The relation Y has three columns P , Q and S . The relation Z has two columns P and T .

X		
P	Q	R
P1	Q1	R1
P2	Q2	R2
P3	Q3	R2

Y		
P	Q	S
P1	Q1	2
P1	Q2	5
P2	Q1	6
P3	Q3	1

Z	
P	T
P1	T1
P3	T2
P4	T3
P4	NULL

Consider the relational algebra expression

$$\Pi_{PRS}[(\sigma_{(Q=Q3 \vee R=R2)}[X \bowtie Y]) \bowtie (\sigma_{(S>1)}[Y \bowtie Z])]$$

where $\bowtie$ denotes natural join operation.

Which of the following options is the correct output for the given expression?

Options A.

One row (P1, R1, 2)

B.

Two rows (P1, R1, 2) and (P1, R1, 5)

C.

Three rows (P1, R1, 2), (P1, R1, 5) and (P2, R2, 6)

D.

Zero rows

Question Type : MCQ

Question ID : 22848210052

Status : Answered

Chosen Option : C

Q.33

Consider the given Python program.

```
def fun(L, i=0):
    if i >= len(L)-1:
        return 0
    if L[i] > L[i+1]:
        L[i+1], L[i] = L[i], L[i+1]
        return 1+fun(L, i+1)
    else:
        return fun(L, i+1)
data = [5, 3, 4, 1, 2]
count = 0
for _ in range(len(data)):
    count += fun(data)
print(count)
```

The output of the program is _____. (Answer in **integer**)

Given 4

Answer :

Question Type : **NAT**Question ID : **22848210068**Status : **Answered****Q.34**

Consider an ER model with the entities $E1(A_{11}, A_{12}, A_{13})$ and $E2(A_{21}, A_{22}, A_{23})$, where, A_{11}, A_{12}, A_{13} are the attributes of $E1$, and A_{21}, A_{22}, A_{23} are the attributes of $E2$. Let A_{22} be a multi-valued attribute. A_{11} and A_{21} are the primary keys of $E1$ and $E2$, respectively.

Let $R12$ be a many-to-many relationship between $E1$ and $E2$. Participation of both $E1$ and $E2$ in $R12$ is total.

The minimum number of relations required to convert the ER model into relational model (assuming there is no other functional dependency) where each relation is in third normal form (3NF) is _____. (Answer in **integer**)

Given 2

Answer :

Question Type : **NAT**Question ID : **22848210071**Status : **Answered**

Q.35

Consider a fully-connected feed-forward multi-layer perceptron. It has 30 neurons in the input layer, followed by two hidden layers and an output layer. The first hidden layer has 4 neurons and the second hidden layer has 3 neurons. The output layer has only one neuron. Assume that no bias parameters are used.

The number of learnable parameters in the multi-layer perceptron is _____.
(Answer in **integer**)

Given 8
Answer :

Question Type : NAT

Question ID : 22848210066

Status : Answered

Q.36

Let $A = \left(I_n - \frac{1}{n} \mathbf{1}\mathbf{1}^T\right)$ be a matrix, where $\mathbf{1} = (1,1,1, \dots, 1)^T \in \mathbb{R}^n$ and I_n is the identity matrix of order n .

The value of $\max_S x^T A x$, where $S = \{x \in \mathbb{R}^n \mid x^T x = 1\}$, is _____.
(Answer in **integer**)

Given 10
Answer :

Question Type : NAT

Question ID : 22848210075

Status : Answered

Q.37

Let there be two relations X and Y as shown. X has three columns P , Q and R . Y has two columns P and S .

X			Y	
P	Q	R	P	S
P1	Q1	R1	P1	10
P2	Q2	R2	P1	15
P3	Q3	R2	P2	20
			P3	1

Consider that the following tuple relational calculus expression is evaluated.

$$\{t \mid t \in X \wedge \exists z \in X (t[P] = z[P]) \wedge \exists m \in Y (m[P] = t[P] \wedge m[S] > 1)\}$$

The number of tuples that will be returned is _____. (Answer in **integer**)

Given 3
Answer :

Question Type : NAT

Question ID : 22848210069

Status : Answered

Q.38

Let $A_{5 \times 5}$ be a matrix such that each of its elements follows *Bernoulli*($p = 0.50$) distribution independently.

The probability that the row-sum of the second row and the column-sum of the third column are both equal to 3 is _____. (Rounded off to **two** decimal places)

Given **0.00**

Answer :

Question Type : **NAT**Question ID : **22848210074**Status : **Answered**

Q.39

Which of the following statements is true for Ridge Regression?

Options A.

The regularizer of Ridge Regression may increase the bias of the model, but it helps in reducing the variance in predictions.

B.

The regularizer in the objective function of Ridge Regression is used to guard against scenarios where the model works well for the test data, but poorly for the training data.

C.

The regularizer of Ridge Regression uses L_1 norm.

D.

Ridge Regression aims to reduce the number of parameters that have negative values.

Question Type : **MCQ**Question ID : **22848210047**Status : **Answered**Chosen Option : **B**

Q.40

Consider a table Employee(EmpID, TeamID), where the column EmpID (ID of an employee) is the primary key. The column TeamID denotes the team ID of the team of which the employee is a member. TeamID is a NOT NULL column.

We want to display the size of the team (denoted as TeamSize) in which each employee is a member by using SQL. As an example, the desired output for the given Employee table is also shown in tabular form.

Which of the following is/are correct?

Employee		Output	
EmpID	TeamID	EmpID	TeamSize
1	8	1	3
2	8	2	3
3	8	3	3
4	7	4	2
5	7	5	2
6	9	6	1

Options A.

```
SELECT E.EmpID, B.TeamSize
FROM Employee AS E, (SELECT TeamID, COUNT(TeamID) AS
TeamSize FROM Employee GROUP BY TeamID) AS B
WHERE E.TeamID = B.TeamID
```

B.

```
SELECT B.EmpID, B.TeamSize
FROM (SELECT EmpID, COUNT(TeamID) AS TeamSize
FROM Employee GROUP BY EmpID) AS B
```

C.

```
SELECT A.EmpID, B.TeamSize
FROM Employee AS A, (SELECT COUNT(TeamID) AS TeamSize FROM
Employee GROUP BY TeamID) AS B
WHERE A.TeamID = B.TeamID
```

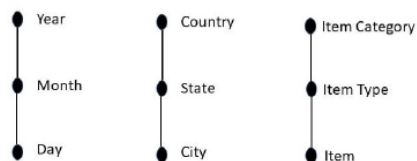
D.

```
SELECT A.EmpID, COUNT(B.TeamID) AS TeamSize
FROM Employee AS A, Employee AS B
WHERE A.TeamID = B.TeamID AND A.EmpID = B.EmpID
GROUP BY A.EmpID
```

Question Type : **MSQ**Question ID : **22848210061**Status : **Answered**Chosen Option : **A,C**

Q.41 Consider the concept hierarchies as shown in the figure.

Which of the following options denotes the total number of possible data cuboids from these concept hierarchies?



Options A.

2^3

B.

$4!$

C.

$2!$

D.

4^3

Question Type : **MCQ**

Question ID : **22848210053**

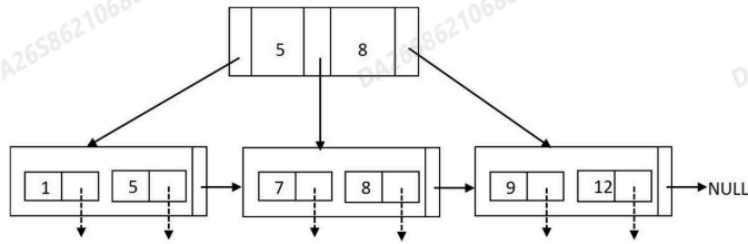
Status : **Answered**

Chosen Option : **A**

Q.42

Consider a B+ Tree where the maximum number of key values in each leaf node is 2 and the maximum number of pointers in each non-leaf node is 3. Let the content of the B+ Tree be as shown in the figure.

Which of the following options denotes the key value(s) stored in the root node after inserting a key value 3 in the given B+ Tree?



Options A.

5

B.

3, 5 and 8

C.

8

D.

3 and 5

Question Type : MCQ

Question ID : 22848210051

Status : Not Answered

Chosen Option : --

Q.43

A clinic specializes in testing for a disease D. The result of the test can be either positive or negative.

A study revealed that if a person suffers from the disease D, the test result in that clinic comes out positive 80% of the time, and negative 20% of the time. If a person is not suffering from the disease D, the test comes out positive 10% of the time and negative 90% of the time. It is also known that among the general population, the disease D occurs in 30% of the individuals.

If a person tests positive for D in that clinic, the probability that he/she actually suffers from the disease D is _____. (Rounded off to **two** decimal places)

Given .24

Answer :

Question Type : **NAT**Question ID : **22848210067**Status : **Answered**

Q.44

Consider the given Python program.

```
def outer():  
    x = []  
    def inner(val):  
        x.append(val)  
        return x  
    return inner  
f1 = outer()  
f2 = outer()  
print(f1(10))    # Line P  
print(f1(20))    # Line Q  
print(f2(30))    # Line R  
print(f1(40))    # Line S
```

Which of the following options is/are correct?

Options A.

Output of Line S is [10, 20, 40]

B.

Output of Line Q is [10, 20]

C.

Output of Line R is [10, 20, 30]

D.

f1 and f2 share the same list x

Question Type : **MSQ**

Question ID : **22848210060**

Status : **Answered**

Chosen Option : **A,B**

Q.45

Assume that a Creative (C) person will Succeed (S) if the person is also Disciplined (D), but will not succeed otherwise. Now, consider the following statements:

- (i) $C \wedge S \Leftrightarrow D$
- (ii) $C \Rightarrow (S \Leftrightarrow D)$
- (iii) $C \Leftrightarrow ((D \Rightarrow S) \vee \neg S)$

Which of the following options is correct?

Options A.

Only (iii) is TRUE

B.

Both (ii) and (iii) are TRUE

C.

Only (ii) is TRUE

D.

Both (i) and (ii) are TRUE

Question Type : **MCQ**

Question ID : **22848210048**

Status : **Not Answered**

Chosen Option : --

Q.46

Let $X_1, X_2, \dots, X_n$ be n independent random variables. Each of the random variables follows $Normal(\mu = 0, \sigma^2 = 1)$ distribution. Define $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$.

Which of the following statements is/are correct?

Options A.

$\sum_{i=1}^n (X_i - \bar{X})^2$ follows Chi-square distribution with $(n - 1)$ degrees of freedom.

B.

$\sum_{i=1}^n X_i^2$ follows Chi-square distribution with n degrees of freedom.

C.

$X_1^2 + X_n^2$ follows exponential distribution with mean 2.

D.

$(\sqrt{n} \bar{X})^2$ follows Chi-square distribution with 2 degrees of freedom.

Question Type : **MSQ**

Question ID : **22848210063**

Status : **Answered**

Chosen Option : **A,D**

Q.47

Consider a directed graph $G = (V, E)$, where V is the finite set of vertices and E is the set of directed edges between the vertices. G may contain cycles but there is no self-loop. Further, G may not be strongly connected.

Let G^R be the graph obtained by reversing the directions of all the edges in G without changing the set of vertices.

Assume that Breadth First Search (BFS) or Depth First Search (DFS) from any given vertex v of a graph visits only the reachable vertices from v in that graph.

Which of the following statements must always be true, regardless of the structure of G ?

Options A.

The order of vertices visited in the BFS of G^R from v is the reverse of the order of vertices visited in the DFS of G from v .

B.

In G^R , the BFS traversal from v will visit exactly the same set of vertices as the DFS from v in G .

C.

If u is a reachable vertex in the DFS of G from v , then v is also a reachable vertex in the BFS of G^R from u .

D.

If u is a reachable vertex in the BFS of G^R from v , then u is also a reachable vertex in the DFS of G from v .

Question Type : MCQ

Question ID : 22848210050

Status : Not Attempted and
Marked For Review

Chosen Option : --

Q.48

Let Account be a relation as shown.

Account	
AccNo	Balance
A1	5000
A2	5000
A3	10000
A4	15000
A5	18000

Consider the given SQL query.

```
SELECT AccNo FROM Account AS A
WHERE (SELECT COUNT(*) FROM Account AS B
WHERE A.Balance < B.Balance) >= (SELECT COUNT(*)
FROM Account AS C WHERE A.Balance > C.Balance)
```

The number of rows returned by the SQL query is _____. (Answer in integer)

Given 1

Answer :

Question Type : NAT

Question ID : 22848210070

Status : Answered

Q.49

Let X be a random variable that follows $Uniform(-1,1)$ distribution. The conditional distribution of the random variable Y given $X = x$ is the $Uniform(x^2 - 0.1, x^2 + 0.1)$ distribution.

The value of $correlation(X, Y)$ is _____. (Answer in integer)

Given 1

Answer :

Question Type : NAT

Question ID : 22848210073

Status : Answered

Q.50 Let

$$L = \lim_{n \rightarrow \infty} \sum_{k=0}^n \frac{e^{-n} n^k}{k!}$$

Which of the following is the value of L ?

Options A.

$$e^{-1}$$

B.

$$0.5$$

C.

$$0$$

D.

$$1.0$$

Question Type : MCQ

Question ID : 22848210055

Status : Not Answered

Chosen Option : --

Q.51

For a given data set $\{x_1, x_2, \dots, x_n\}$, where $n = 100$, it is known that

$$\frac{1}{2000} \sum_{i=1}^n \sum_{j=1}^n (x_i - x_j)^2 = 99$$

Let us denote $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$ The value of $\frac{1}{99} \sum_{i=1}^n (x_i - \bar{x})^2$ is _____. (Answer in integer)

Given 1

Answer :

Question Type : NAT

Question ID : 22848210072

Status : Answered

Q.52

Let $P(x)$ be a predicate.

Which of the following statements is/are **NOT** valid in first-order logic?

Options A.

$\forall x P(x) \Rightarrow \exists x \neg P(x)$

B.

$\exists x P(x) \Rightarrow \forall x P(x)$

C.

$\forall x P(x) \Rightarrow \exists x P(x)$

D.

$\exists x P(x) \Leftrightarrow \forall x P(x)$

Question Type : **MSQ**

Question ID : **22848210058**

Status : **Answered**

Chosen Option : **A**

Q.53

A recursive function in Python is given.

```
def mystery(n):  
    if n <= 0:  
        return 1  
    else:  
        return mystery(n-1) + mystery(n-2)
```

Now, consider the following function call:

mystery(4)

Assume that a typical runtime stack is used to manage function calls. Each function call is pushed onto the stack and removed only after it finishes execution.

Which of the following options denotes the total number of function calls (i.e., the total number of stack activations), including the initial call, to compute mystery(4)?

Options A.

17

B.

9

C.

5

D.

15

Question Type : MCQ

Question ID : 22848210049

Status : Not Answered

Chosen Option : --

Q.54

Consider that 20 stories of Author X and 10 stories of Author Y were kept together without mentioning the names of the authors. A classifier was then asked to predict the author (X or Y) of each of these stories. Let, out of X's stories, 6 were classified as that of Y. On the other hand, out of Y's stories, 2 were classified as that of X.

Considering X and Y as two classes, which of the following statements is/are true?

Options

- A. Recall of Class X is higher than the Recall of Class Y.
- B. Precision of Class X is higher than the Precision of Class Y.
- C. Accuracy of the classifier is 14/15.
- D. Accuracy of the classifier is 11/15.

Question Type : MSQ

Question ID : 22848210057

Status : Answered

Chosen Option : A,D

Q.55

Let X and Y be two independent random variables. X follows $Bernoulli(p = 0.3)$ distribution and Y follows $Normal(\mu = 0, \sigma^2 = 100)$ distribution.

Which of the following options is the variance of $(2X - 1)Y$?

Options A.

90

B.

21

C.

100

D.

49

Question Type : MCQ

Question ID : 22848210054

Status : Not Answered

Chosen Option : --